THE TYPE OF A FORMAL ALGEBRAIC SPACE CAN BE CHECKED ON ANY ONE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's parenthetical '(small detail omitted on finite disjoint unions of affine formal algebraic spaces)'; it is reproduced verbatim and recorded as an obligation of the card.

Editorial note. The proof displays one coproduct covering the given object; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 7 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The definition of a formal algebraic space through a presentation. Same chapter, Definition definition-formal-algebraic-space, invoked at proof lines 4694-4695 and 4701-4705. Fixes the presentations used in both formulations.
- Fibre products of formal algebraic spaces along a representable diagonal are formal algebraic spaces, so coverings may be chosen. Same chapter, Lemma lemma-fibre-products-general, invoked at proof lines 4702-4703. Produces the coverings from which the finite refinement is selected.
- The type ascends along a representable etale affine morphism. Same chapter, Lemma lemma-property-goes-up-affine, invoked at proof lines 4713-4716. Transfers the type to the selected members.
- The two-way statements for the five types. Same chapter, Lemmas lemma-iff-countably-indexed, lemma-iff-cic, lemma-iff-weakly-adic, lemma-iff-adic-star and lemma-iff-noetherian, invoked at proof lines 4718-4729. Conclude the equivalence for each type.

Source reference. Original lemma 0AJ2: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3488–3488. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKT: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4642–4642. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GXR: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4597–4597. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4549–4549. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKW: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4655–4655. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GXS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4628–4628. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKQ: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4373–4373. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is a *formal algebraic space* if there exist a family of maps $\{X_i \to X\}_{i\in I}$ of sheaves such that

(1) X_i is an affine formal algebraic space,
(2) $X_i \to X$ is representable by algebraic spaces and étale,
(3) $\coprod X_i \to X$ is surjective as a map of sheaves

and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let*

$$P \in \left\{ \begin{array}{c} \text{countably indexed}, \\ \text{countably indexed and classical}, \\ \text{weakly adic, adic}*, \text{ Noetherian} \end{array} \right\}$$

Let X be a formal algebraic space over S. The following are equivalent

(1) *if Y is an affine formal algebraic space and $f : Y \to X$ is representable by algebraic spaces and étale, then Y has property P,*
(2) *for some $\{X_i \to X\}_{i\in I}$ as in Definition 0.1 each X_i has property P.*

Proof. It is clear that (1) implies (2). Assume (2) and let $Y \to X$ be as in (1). Since the fibre products $X_i \times_X Y$ are formal algebraic spaces (Lemma 0AJ2) we can

pick coverings $\{X_{ij} \to X_i \times_X Y\}$ as in Definition 0.1. Since Y is quasi-compact, there exist $(i_1, j_1), \ldots, (i_n, j_n)$ such that

$$X_{i_1 j_1} \amalg \ldots \amalg X_{i_n j_n} \longrightarrow Y$$

is surjective and étale. Then $X_{i_k j_k} \to X_{i_k}$ is representable by algebraic spaces and étale hence $X_{i_k j_k}$ has property P by Lemma 0AKQ. Then $X_{i_1 j_1} \amalg \ldots \amalg X_{i_n j_n}$ is an affine formal algebraic space with property P (small detail omitted on finite disjoint unions of affine formal algebraic spaces). Hence we conclude by applying one of Lemmas 0AKS, 0GXR, 0GXS, 0AKT, and 0AKW. □